Uncertainty Communication for Decision-making

CMSC839E: Advanced Topics in Human-Computer Interaction

September 2, 2025 @ UMD by Fumeng Yang https://fumeng-yang.github.io/CMSC839E

Fumeng Yang

foo-mung

Fumeng

foo-mung

Hmm...

Me: HCI + Visualization

Assist. Prof.	🐢 UMD CS
Postdoc	🐈‍⬛ Northwestern University CS
PhD	🐻 Brown University CS
Masters	🐘 Tufts University CS
personal	💁‍♀️ **www.fmyang.com**
lab	**fig-x.github.io**

We have a TA!! ⭐

Zhongzheng Xu

Teaching Assistant

Pronouns: he/him

Activity

Raise your hands if you are in a PhD program

Raise your hands if you are in a Master program

Raise your hands if you are in your first year of grad school

Have first-author papers at a major conference/journal

Have done peer reviews for a conference or journal

Uncertainty Communication for Decision-making

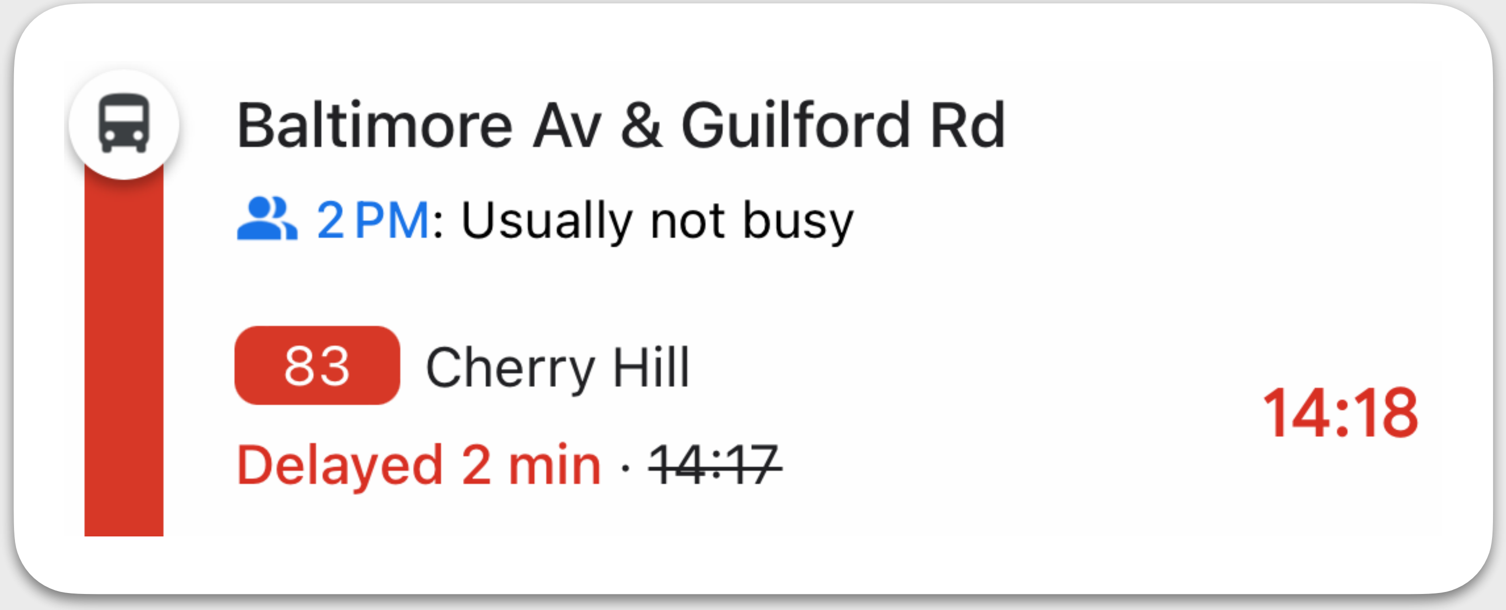
CMSC839E: Advanced Topics in Human-Computer Interaction

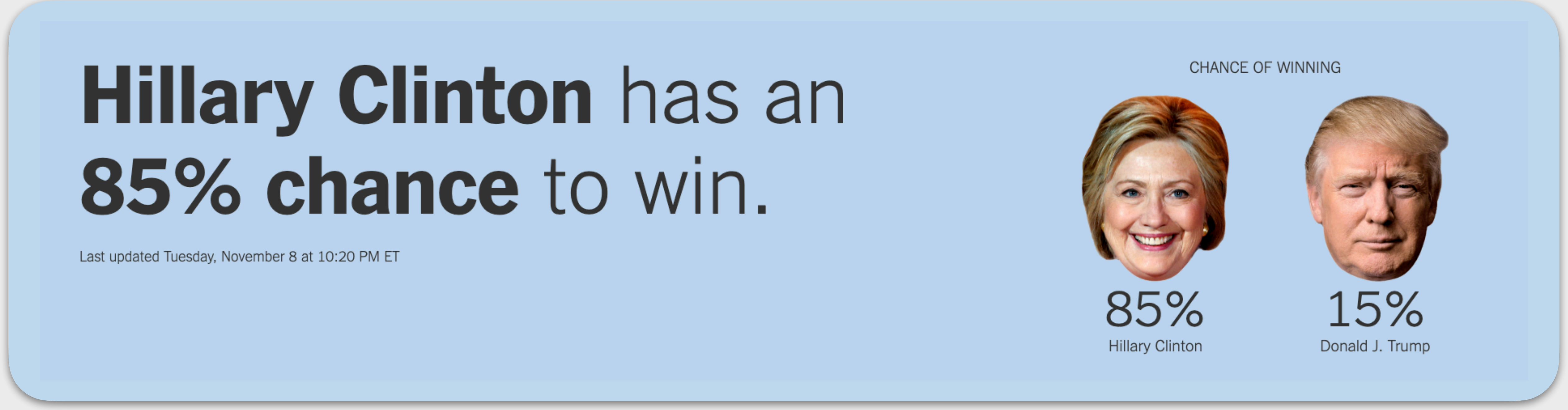
What do you think this course is about?

The big picture

Vision

We are surrounded by many computational models. → LLM, election forecasts, traffic time estimate, recommenders, …





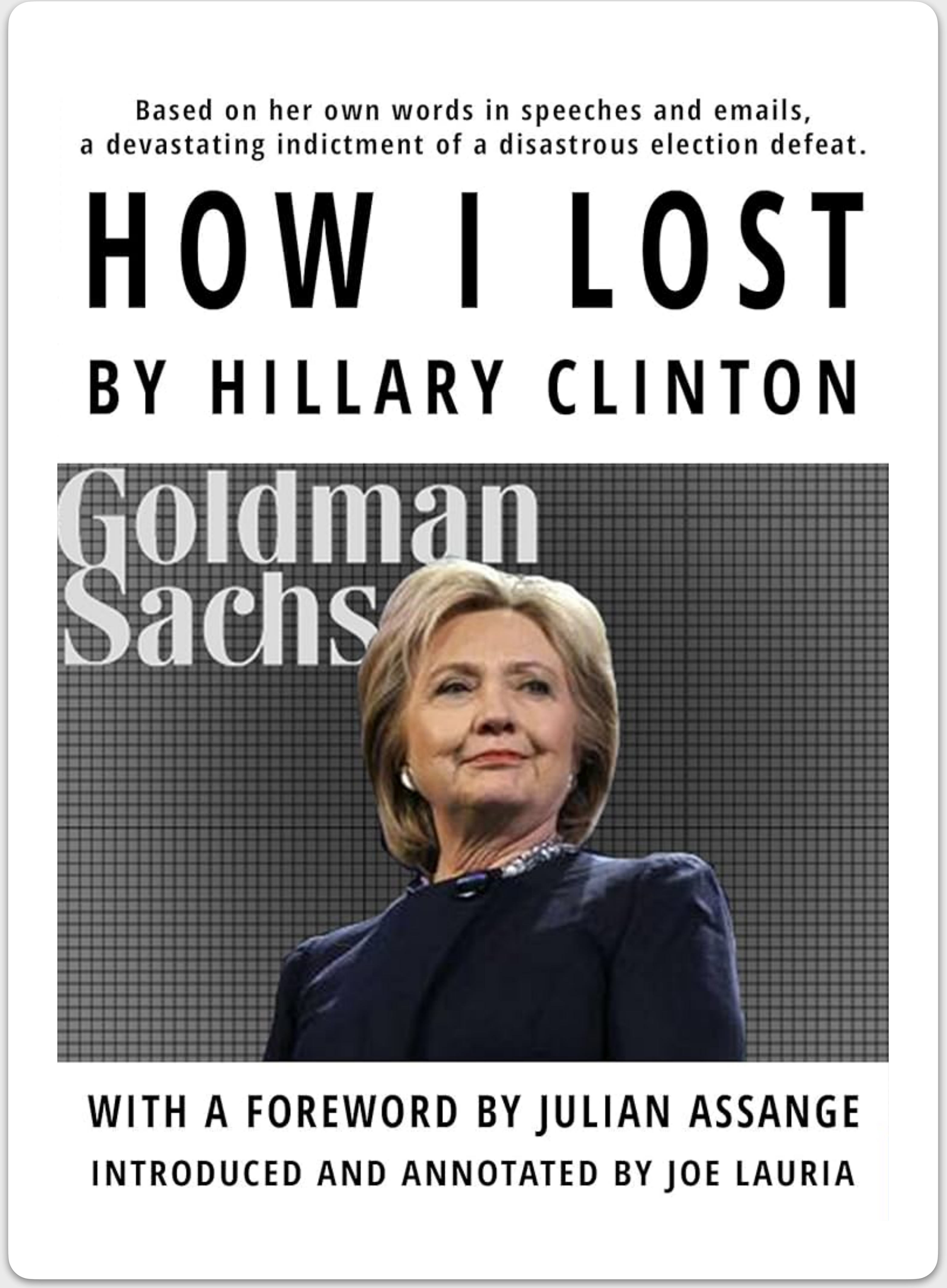
Gemini

https://www.techradar.com/computing/artificial-intelligence/best-llms

Vision

We are surrounded by many computational models.

All models are imperfect.



Vision

We are surrounded by many computational models.

All models are imperfect. → miss a bus/a meeting, medical decisions, voting, …

How to cope with these imperfections for human users to make the right decision? → individual decision, public policy, election results, …

Vision

We are surrounded by many computational models. → LLM, election forecasts, traffic time estimate, recommenders…

All models are imperfect. → uncertain, wrong, hallucination,…

How to cope with these imperfections for human users to make the right decision? → individual decision, public policy, election results, …

Vision

We are surrounded by many computational models.

All models are imperfect.

How to cope with these imperfections for human users to make the right decision?

data

architecture

training

user

touch on

this course

in light of uncertainty

Course scope

HCI

ML/AI

LLM

psy

Human-AI

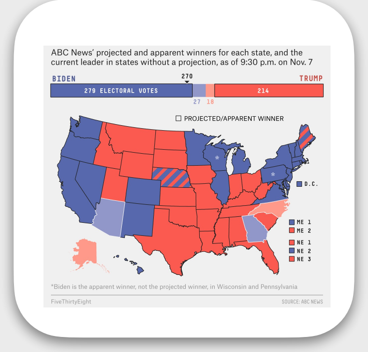
stats

edu

ethics

VIS

context



Course goals

Not a hardcore ML/AI/LLM/Statistics course.

I hope you can improve your understandings of these. We welcome different opinions.

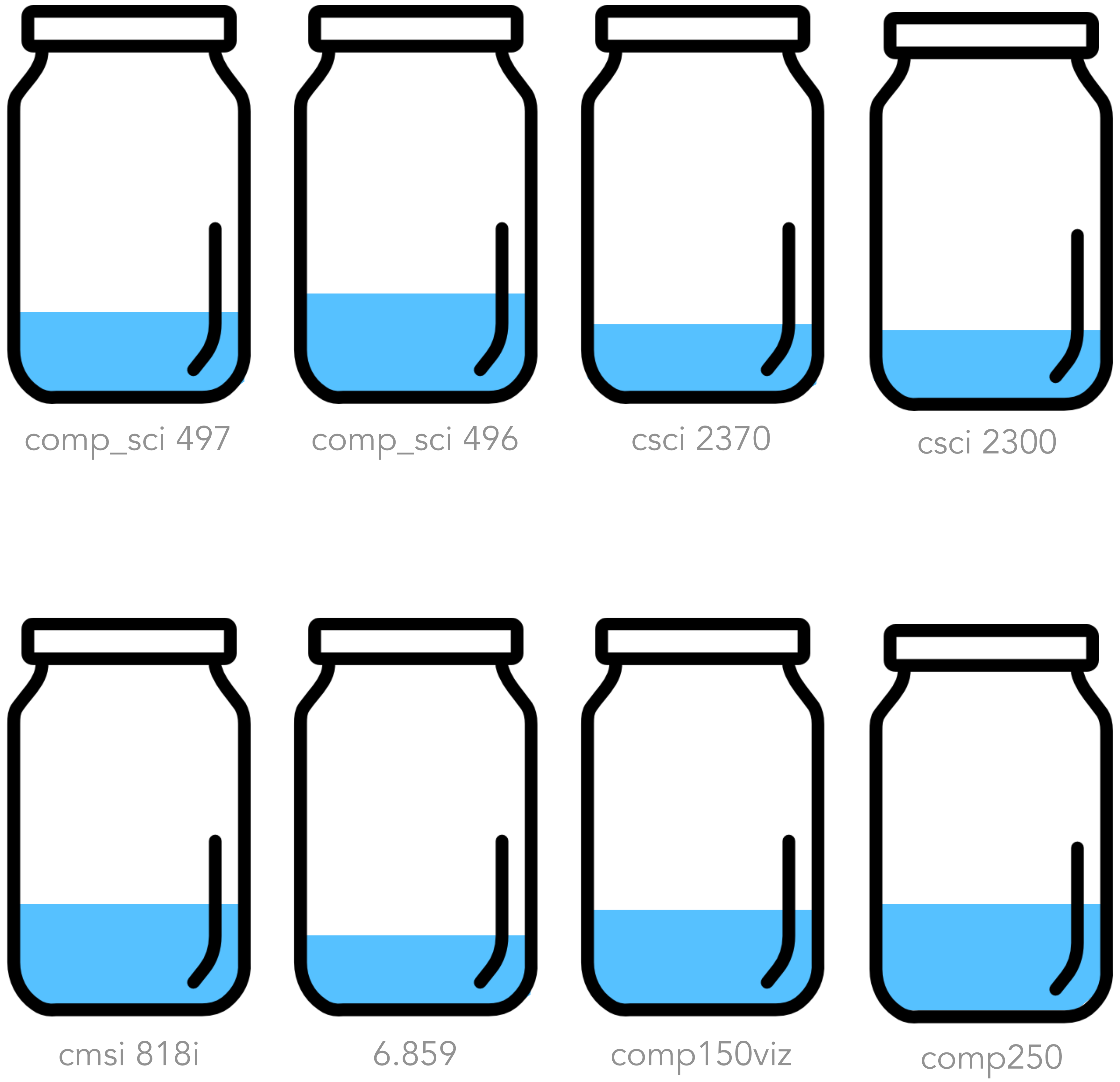
Low programming but it depends on the project of your choice.

I assume you have some programming skills?

Train junior grads on **how to do HCI research.**

quantitative analysis, paper writing, reviewing, project management,
tho your advisor may have a different opinion; MS for grad schools & useful methods

How I made this course



Taking the part I like about from those courses I took, attended, or taught.

I hope this will be a **like-able** and **useful** course.

I think my teachers'd be happy to see that I pass down their wisdom to the next generation.

And I wish (some of) you can do so as well one day.

Questions?

Course organization

Course website & Canvas

https://fumeng-yang.github.io/CMSC839E/

Course materials will be publicly released on this website.

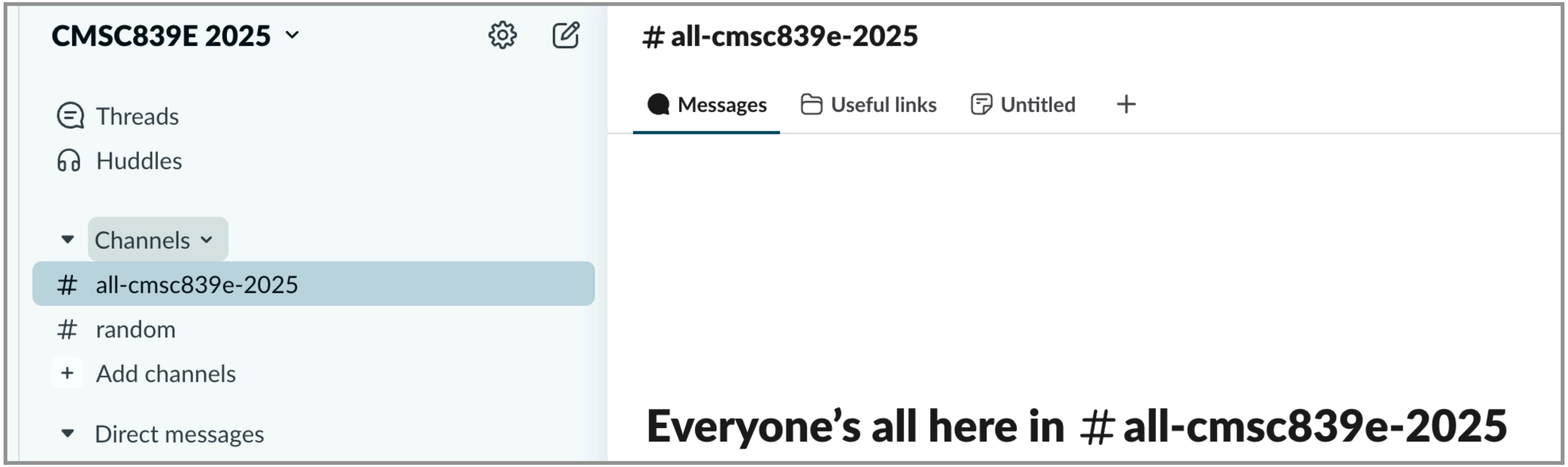
Confidential information will be class-only (e.g., on UMD Google Drive).

The webpage is embedded on Canvas.

We are on Slack!

We should have added you all.

There is an invite link on Canvas.



Canvas = a hub!

Fall 2025

- Home
- **Syllabus**
- Assignments
- Grades
- People
- Collaborations
- Chat

CMSC839E-0101,PJ01: Advanced Topics in Human-Computer Interaction; Uncertainty Communication for Decision-Making-Fall 2025 fmy

Jump to Today

Course staff and hours

Please use Slack for questions; we don't reply to emails.

Fumeng Yang

Assistant Professor
Pronouns: she/her
Hours: W after class to 4:30 PM
IRB 5138

Zhongzheng Xu

Teaching Assistant
Pronouns: he/him
Hours: Wed 1-2 PM
https://umd.zoom.us/j/99360911431

Presentation and Role Play Signup | **Slack Invite** | **Emotion Share Form**

Course website: https://fumeng-yang.github.io/CMSC839E, embedded below.

CMSC839E

Advanced Topics in Human-Computer Interaction (Fall 2025)

Uncertainty Communication for Decision-making

24/7 Canvas Chat Support

....or call 1-833-566-3347 (staff/faculty)
1-877-399-4090 (students)

- View Course Stream
- View Course Calendar
- View Course Notifications

To Do

- Sep 15, 2025 (Mon)
  CMSC839E-0101,PJ01: Advanced Topics in Human-Computer Interaction; Uncertainty Communication for Decision-Making-Fall 2025 fmy
  Sep 15 at 7am
- Sep 17, 2025 (Wed)
  CMSC839E-0101,PJ01: Advanced Topics in Human-Computer Interaction; Uncertainty Communication for Decision-Making-Fall 2025 fmy
  Sep 17 at 7am
- Project proposal
  CMSC839E-0101,PJ01: Advanced Topics in Human-Computer Interaction; Uncertainty Communication for Decision-Making-Fall 2025 fmy
  5 points |

Course topics

[Demo]

Topics are not in a learning order…

Grading & How this works

Grading

15% Weekly reading responses	→	questions for papers
18% Class presentations	→	paper and project presentations
5% Course attendance	→	in a minute
32% Two assignments	→	course project/review
15% Midterm exam: take-home 15% Final exam: revise your paper	→	MS/PHD qualifying
10% Extra credit	→	exceptional performance

Grading

15% **Weekly reading responses◀**

18% Class presentations

5% Course attendance

32% Two assignments

15% Midterm exam: take-home

15% Final exam: revise your paper

10% Extra credit

Weekly reading responses

1-2 questions for each required paper by AoE prior to the class.

≥1 questions per class should be non-trivial.

We **won't** strictly count non-trivial questions.

Show us your efforts and your critical thinking.

Lateness: …try to submit on time because they'll be used in class.

Grading

15% Weekly reading responses

18% **Class presentations** ◀

5% Course attendance

32% Two assignments

15% Midterm exam: take-home

15% Final exam: revise your paper

10% Extra credit

Class presentations - 18%

8% role play: 1 paper presentation

5% role play: other roles twice

5% 1 group presentation of your course project

Three role plays **shouldn't be** on the same day or for the same paper…

Paper presentations (8-12 minutes) - 8%

🧑‍💼 Orator

Summarize its main methods, findings, and takeaways clearly and engagingly.

🧑‍🏫 Tutor

You are a teacher providing a step-by-step tutorial showing how to use the paper's tools, datasets, or methods.

Other roles (3-5 minutes per role) - 5%

👷 Digger

Where this paper sits in the context of previous and subsequent work

🧑‍🏫 Detective

Background check on the paper's authors

🧑‍🎓 Researcher

Propose an imaginary follow-up project

🧑‍⚖️ Peer Reviewer

Present a brief review of the paper

Most classes...

1-3m | 8-12m | 10-15m

👩 **Me** → 🧑‍💼 **Orator** or 🧑‍🏫 **Tutor** → 👷 **Digger**, 🕵️ **Detective**, 🧑‍🎓 **Researcher**, 🧑‍⚖️ **Peer Reviewer**

x 2

→ ~15m discussion on questions submitted

Please use Slack to coordinate…

Presentation signup

Find link on the course website & Canvas; start from the 3rd week

Use your UMD email; Slack me your email if you can't access

We freeze signup on Monday for the next week (unless no signup…)

We freeze presenters and roles for next week on the Monday prior to that week (except for no signup)									
Week	**Date**	**Topic**	**Papers**	**Slide de**	**Main presenter(s)**	**Digger**	**Detective**	**Researcher**	**Peer Reviewer**
week 1	9/1/2025	Labor Day (no class)							
	9/3/2025	Introduction	Presentation & Role Play						
week 2	9/8/2025	Ice breaker: bring a chart I	Bring a chart I						
	9/10/2025	Ice breaker: bring a chart II	Bring a chart II						
week 3	9/15/2025	AI & Bias - Model	[R1] Cognitive...						
			[R2] AI Can Be...						
	9/17/2025	AI & Bias - Human	[R3] Lessons...						
			[R4] Deciding...						
week 4	9/22/2025	LLM & Uncertainty	[R5] From Calibration...						
			[R6] "I'm Not Sure" ...						
	9/24/2025	Hallucination & Misinformation	[R7] Fakes...						
			[R8] Trust It...						
week 5	9/29/2025	Human-AI Decision-Making I	[R9] Rethinking...						
			[R10] Utilizing...						
	10/1/2025	Project Discussion							

One slide deck for one paper

Grading

15% Weekly reading responses

18% Class presentations

5% **Course attendance** ◀

32% Two assignments

15% Midterm exam: take-home

15% Final exam: revise your paper

10% Extra credit

Course attendance - 5%

No attendance requirement

= sad attendance 🥲

= me & other students 💔 in a cold classroom 🥶

Life happens. At least 60-70% of the time. Don't send me excuses.

No check-in sheet. Take some (fun) pictures to show you are here.

~10 photos from different classes throughout the semester 👍

Questions?

Grading

15% Weekly reading responses

18% Class presentations

5% Course attendance

32% **Two assignments** ◀

15% Midterm exam: take-home

15% **Final exam: revise your paper** ◀

10% Extra credit

Assignment P: Paper & project - 21%

Assignment R: Peer review - 11%

Final exam: Address peer reviews - 15 %

Assignment P: Paper & project - 21%

The major major thing of this course (2-3 people)

P.1 write a proposal using the template at week 5 - 5%

P.2 abstract & author information at week 11 - 1%

P.3 paper submission at week 12 - 15%

Assignment P: Paper & project - 21%

The major major thing of this course (2-3 people)

start to think about it now

P.1 - have your team at week 5 - 5%

… do the work

P.2 - have some results at week 11 - 1%

… write the paper

P.3 - final paper at week 12 - 15%

presentation

Assignment P: Paper & project - 21%

The major major thing of this course (2-3 people)

P.1 - have your team at week 5

P.2 - have some results at week 11 ← not submitting on time = 0 of P.3

P.3 - final paper at week 12 ← no late submission accepted

Assignment P: Paper & project - 21%

(Loosely) align with the topics discussed in this course.

Our course topics are broad enough to allow almost any HCI/LLM/Vis/CS projects. You are encouraged to explore intersections with your own research interests or field of study.

AI/ML/NLP: e.g., an LLM + bias/debias/explanation/uncertainty/decision
HCI or psychology: e.g., human + LLM/AI/model/uncertainty/trust/vis/decision project
other field: make a connection to LLM/Vis/uncertainty
Something like people's perception of different print papers - out of scope

Assignment R: Peer Review - 11%

Look at each other papers and write 1 review (solo)

R.1 - bidding & conflict

Indicate your preference

R.2 - write reviews for other papers

Final exam: Address peer reviews - 15 %

Revise your paper based on the peer reviews (group)

Assignment R + final exam

To simulate the paper submission process

Can’t fully simulate the anonymous part.

I look for **reasonable** reviews, and you aim for **helpful** reviews.

I will read all to ensure they’re reasonable.

When grading final exams, I also look for **reasonable** responses

PR + final exam + presentation = a project cycle

- Brainstorm & discussion
- Do the work
- Analyze the data
- Submit an abstract
- Submit the paper
- Get reviewed
- Revise your paper
- Accepted & present at conference

Due to course planning,

we have to joggle things around…

PR + final exam + presentation = a project cycle

CHI
VIS
CSCW
Ubicomp
VR
Journal…

Brainstorm & discussion

Do the work

Analyze the data

Submit an abstract

Submit the paper

Get reviewed

Revise your paper ?

Accepted & present at conference

IUI/UIST/
SIGGRAPH/most
AI conferences

Rebuttal
proposed changes

QPR + final exam + presentation = a project cycle

Brainstorm & discussion

Do the work

Analyze the data

Submit an abstract

Submit the paper

Get reviewed

Revise your paper

Accepted & present at conference

Rejected

Questions?

Grading

15% Weekly reading responses

18% Class presentations

5% Course attendance

32% Two assignments

15% **Midterm exam: take-home ◀**

15% Final exam: revise your paper

10% Extra credit

Midterm exam

We select ~15 questions for all questions discussed in class.

You select 7 questions to answer. 1-3 paragraphs for each

AI usage

In line with the spirit of this course…

- AI should **not** be used to **directly complete** coursework.
- Clearly **mark** any content that is directly generated by AI.
- Be **mindful** about your trust in AI-generated content.
- You may use AI to **assist** with coursework.
- We encourage you to use AI **creatively** and share your experiences.
- Violations may result in a zero for the affected portion.

Miscs

Lateness: things like abstracts and (revised) papers must be on time

1-2 days extension if you have a good reason for others

Questions expire after class so please on time, skip 1 submission = little affect

Office hours: Wed after class to ~4:30pm

Office: IRB 5318

Location of office hours: classroom to my office

Communication: use Slack

Questions?

Other things

How to read a paper

How long it takes me to read a paper…

Reviewing a paper: 3-8 hrs

.25 hrs

If I want the gist

.5 - 2 hrs

My expectation, 4-6 hrs / week + projects

12+ hrs

If I'm editing a student's first draft

How I read an HCI paper

(Find their presentation/demo video)

Definitely abstract, introduction

Methods

Results (it depends)

Related work (maybe)

Discussion (sometimes they are fun to read)

Conclusion (I almost never read it, but some people put this in the first line)

Questions?

Next week

Next week: icebreaking, bring a chart

Start to read papers and sign up for presentations

To be continued…